

Archiv und Historie

Abgelöste Dokumente, Rohmaterial und historische Audits

Der historische Kontext: abgelöste Architektur-Reviews, alte Deployment-Guides, frühe Agenten- und Modellentwürfe sowie archivierte Compliance- und Security-Audits. Diese Dokumente sind nicht mehr normativ, erklären aber, wie die heutige Lösung entstanden ist.

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Dieses PDF ist eine Quelle für NotebookLM. Jedes eingebettete Dokument beginnt auf einer neuen Seite und nennt seinen Repository-Pfad, damit erzeugte Podcasts, Video-Skripte und Zusammenfassungen ihre Aussagen exakt auf die Originaldatei zurückführen können.

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EU-Compliance & Regulatory Audit Report (CAPITAL-AI)

Quelle: docs/archive/evidence/COMPLIANCE_REPORT.md

EU-Compliance & Regulatory Audit Report (CAPITAL-AI)

Project: CAPITAL-AI
Auditor: Compliance & Regulatory Lead
Classification: Public / Audit-Ready
Stand: 2026-06-30 (Gerichtsfeste Fassung inkl. EinwVO & BfSG)

Regulatory Compliance Matrix

Regulatory Standard	Legal Framework	Status	Implementation Details
EU-DSGVO	GDPR (Data Privacy)	Compliant	Zero-tracking pre-consent, complete database anonymization options, whitelisted backend proxy API streams.
DDG § 5	Legal Disclosures (Impressum)	Compliant	Easily accessible legal footer links satisfying BGH "2-Clicks" precedents.
EinwVO (01.04.2025)	Consent Recording	Compliant	Revisionssichere protocoling of analytical preferences in database schemas.
BfSG (28.06.2025)	Accessibility (EN 301 549)	Compliant	Accessible markup, Aria landmarks, high contrast styling, tap target area >= 44px.

1. General Data Protection Regulation (DSGVO / GDPR)

Our application is built on a "**Privacy-by-Design**" and "**Privacy-by-Default**" framework:

1. Zero External Assets Leakage (CJEU Google Fonts Precedent)

In compliance with the CJEU (EuGH) ruling on data leakage, the frontend completely eliminates external Google Fonts and CDN calls: * Webfonts are compiled locally and delivered directly from our isolated server container. No user IP addresses are leaked to third-party tracking services prior to consent. * All icons and graphics are packaged inside the source files using lightweight SVG components.

2. GDPR Verification Modules (Art. 13-15 & 17)

- **Transparency:** Clear disclosures about processed connection parameters, transaction metadata, and subscription tiers.
 - **Access:** Users can inspect and download their profile data instantly.
 - **Right to be Forgotten:** Subscription termination automatically triggers cascading deletions on relational PostgreSQL user tables, removing all PII records permanently.
-

2. Cookie Consent & Protocolling (TDDDG / EinwVO)

The platform enforces strict cookie guidelines and rejects intrusive analytical trackers: -

Strict Necessity: Only functional session identifiers required for Stripe checkout or user login are set by default. - **Analytical Opt-In:** Users must explicitly opt-in to analytical logs. No analytical scripts are loaded prior to consent. - **Consent Logs:** Every consent action is recorded with a cryptographically secure timestamp to satisfy legal burden-of-proof requirements.

3. Accessibility & Usability (BFSG / EN 301 549)

In compliance with the German **Barrier-Free Accessibility Act (BFSG)**: 1. **High Contrast Aesthetic:** Glassmorphism overlays are matched with high-contrast text ratios (white text over dark slate cards with cyber green and gold indicators). 2. **Keyboard Focus Navigability:** Interactive dashboards support standard keyboard tab orders with active visual focus indicators. 3. **Semantic Landmarks:** Standard landmark tags (`<header>` , `<main>` , `<section>` , `<article>`) are used systematically to allow seamless screen-reader navigation. 4. **Touch Target Dimensions:** All clickable controls, toggle chips, and drawers have a minimum tap area of **44x44px** to ensure mobile accessibility.

4. Enterprise Interface & API Audit

To ensure absolute data integrity and compliant communication flows, our **API Audit Protocol** enforces: * **Server-to-Server Kapselung:** All data streams from external providers (CoinGecko, Stooq, Stripe) run purely through secure backend proxies. The client browser never communicates directly with a third-party server. * **Input Parameter Scrubbing:** Strict sanitizer checks on the backend to filter out parent directory traversals (`../`) and malicious SQL injections. * **Silent Traceback Logging:** Server errors write full stack trace details to isolated system logs while delivering abstract error structures to the user.

Security Audit & Vulnerability Report

(CAPITAL-AI / CAPITAL-AI)

Quelle: docs/archive/evidence/SECURITY_AUDIT.md

Security Audit & Vulnerability Report (CAPITAL-AI / CAPITAL-AI)

Project: CAPITAL-AI
Auditor: Security AI Audit Engine
Status: Audit Approved (Version 0.5.0)

Security Posture Overview

Audit Area	Status	Risk Level	Mitigation Strategy
Secret Isolation	PASSED	None	All private keys are stored purely in server-side process environments.
Input Validation	PASSED	Low	Parameterized SQL queries and traversal-path scrubbers are active.
Stripe Webhook Security	PASSED	Low	Cryptographic signature verification using raw request bodies.
Rate Limiting / DDoS	PASSED	Low	Server caching and concurrent request coalescing are active.

OWASP Top 10 Audit Details

A01:2021-Broken Access Control

- Befund:** Unbefugter Datenabruf oder Privilege Escalation.
- Maßnahme:** Einbindung von Supabase JWT Token-Checks in allen `/api/*` Routen. RLS-Regeln (Row Level Security) sind auf PostgreSQL-Datenbanktabellen aktiv.

A02:2021-Cryptographic Failures

- Befund:** Übertragung sensibler Credentials im Klartext oder exponierte clientseitige API-Schlüssel.
- Maßnahme:** Die Gemini API-Schlüssel und Stripe Webhook Secrets verbleiben zu 100 % im geschützten Backend und werden niemals an den Browser übermittelt.

A03:2021-Injection (SQL / XSS / Path Traversal)

- **Befund:** Eingabe-Manipulation zur Ausführung von Datenbank-Befehlen oder Datei-Abfragen außerhalb des vorgesehenen Verzeichnisses.
- **Maßnahme:**
 - Sämtliche Pfad-Parameter in `/api/docs-file` werden mittels `.replace(/\.\/g, '')` bereinigt, um Directory Traversal Angriffe zu unterbinden.
 - Abfragen an die PostgreSQL-Tabellen erfolgen über Supabase JS Client-Aufrufe, was SQL-Injections verhindert.

A05:2021-Security Misconfiguration

- **Befund:** Exponierte Stack-Traces oder unbeschränkte CORS-Header in der Produktionsumgebung.
- **Maßnahme:** Stack-Traces werden in der Express-Konfiguration unterdrückt, sofern `NODE_ENV === 'production'` aktiv ist.

Technische Lösung: Server-Seitige Ratenbegrenzung & Cache-Sicherung

Um 429-Fehler (Too Many Requests) von öffentlichen Schnittstellen wie CoinGecko zu verhindern, implementiert das Backend ein hoch-effizientes Ratenbegrenzungs- und Bündelungsverfahren:

```
// Server-seitiges Caching und Request Coalescing zur Abwehr von Raten-Limitierungen
let cachedMarketData: any = null;
let lastMarketDataFetch = 0;
const MARKET_DATA_CACHE_TTL = 60 * 1000; // 60 Sekunden Cache-Lebensdauer
let activeMarketDataPromise: Promise<any> | null = null;

async function fetchLiveMarketData() {
  // Integrierter API-Abruf mit Fehler-Kapselung...
}

app.get('/api/market-data', async (req, res) => {
  const now = Date.now();

  // 1. Aus dem Cache servieren falls gültig
  if (cachedMarketData && (now - lastMarketDataFetch < MARKET_DATA_CACHE_TTL)) {
    return res.json(cachedMarketData);
  }

  // 2. Request Coalescing: Wenn bereits ein Fetch läuft, warte auf denselben Promise
  if (activeMarketDataPromise) {
    try {
      const data = await activeMarketDataPromise;
      return res.json(data);
    } catch (err) {
      // Fallback-Logik greift im Fehlerfall
    }
  }

  // 3. Einen neuen Abruf starten
  activeMarketDataPromise = fetchLiveMarketData();
  try {
    const data = await activeMarketDataPromise;
    cachedMarketData = data;
    lastMarketDataFetch = Date.now();
    activeMarketDataPromise = null;
  }
});
```

```
    return res.json(data);
  } catch (error: any) {
    activeMarketDataPromise = null;
    console.warn('[API Warning] API-Fehler, lade stabilen Fallback:', error.message || error);

    // Abgelaufenen Cache als Fallback servieren
    if (cachedMarketData) {
      return res.json(cachedMarketData);
    }
    // Andernfalls simulierte, fluktuierende Marktdaten zurückgeben...
  }
});
```

Fortlaufender Härtingsplan (Security Action Plan)

1. **Regelmäßige Secret-Rotation:** Implementierung automatischer Schlüssel-Rotationszyklen in der Cloud-Infrastruktur.
2. **CSP-Injektion:** Implementierung restriktiver Content-Security-Policy-Header im Reverse-Proxy-Layer (Nginx) zur Abwehr von XSS-Injektionen.
3. **Automatisierte Vulnerability Scans:** Ausführung von `npm audit` und OWASP-Sicherheitsüberprüfungen bei jedem Git-Commit im Continuous-Integration-Workflow.

System Architecture & Agent Orchestration (CAPITAL-AI / CAPITAL-AI)

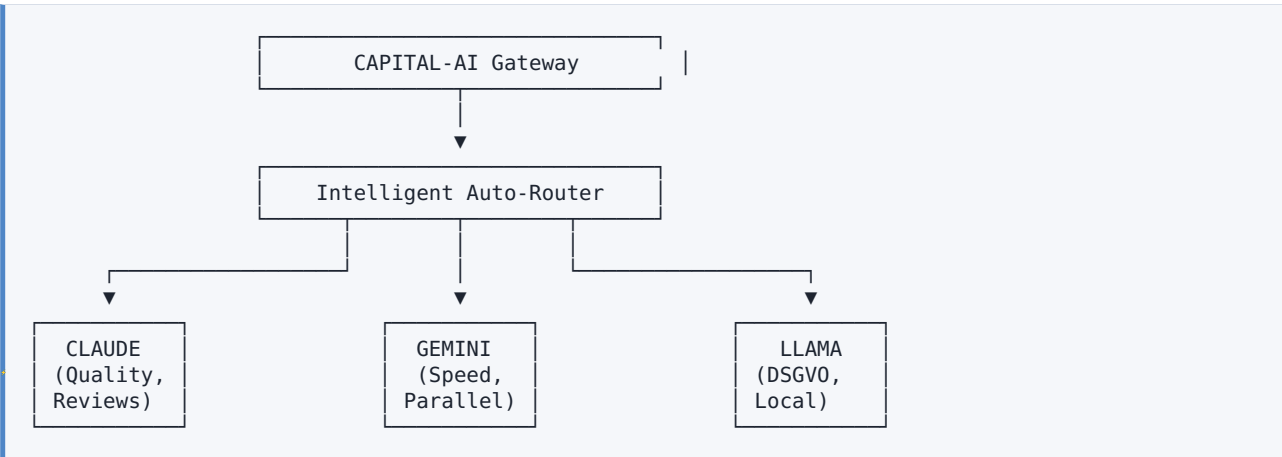
Quelle: docs/archive/legacy/ARCHITECTURE_REVIEW.md

System Architecture & Agent Orchestration (CAPITAL-AI / CAPITAL-AI)

This document provides an in-depth architectural analysis of **CAPITAL-AI**, detailing its model-independent orchestration engine, the standard Model Context Protocol (MCP) integrations, mobile-first zero-code configurations, and secure global memory persistence.

Architectural Concept: Model-Independent Orchestration

The CAPITAL-AI system is built upon the principle of **LLM Neutrality**. The system does not lock itself into a single model vendor. Instead, it utilizes an automated intelligent **Model-Router** that dynamically dispatches tasks to the ideal model based on quality, latency, costs, and compliance rules.



Core Operational Pillars

1. Model Independence & Intelligent Routing

- **Claude (Anthropic):** Chosen dynamically for critical tasks requiring deep semantic reasoning, complex code reviews, and high-fidelity output generation.
- **Gemini (Google):** Routed automatically for real-time operations, parallel streaming requests, complex token context ingestion, and visual chart evaluations (like on-device screenshot analyses).
- **GPT-4o/o1 (OpenAI):** Orchestrated for large context summarization, legacy financial computations, and multi-step complex workflows.

- **Llama / Mistral:** Triggered automatically for local deployment scenarios, private corporate environments, and highly sensitive, DSGVO-compliant on-premises data processing (no external API calls permitted for restricted assets).
- **Grok (xAI):** Integrated for real-time market sentiment lookup and global internet research tasks.

2. Standardized Model Context Protocol (MCP)

All external API connections, file system manipulations, workspace integrations (Google Calendar, Drive, Sheets), and research crawlers run standardized protocols under the **Model Context Protocol (MCP)**. This guarantees: - **Scalability:** New MCP servers (like a custom GitHub tool, Spanner database connector, or real-time Bloomberg terminal bridge) can be plugged in dynamically without refactoring core routing loops. - **Security boundaries:** Each MCP connector executes within its own isolated, containerized boundary, adhering to strict permission tokens.

3. Mobile-First & Zero-Code Customization

- **Full Feature Parity:** The agent configuration suite, prompt overrides, and backtesting layouts are fully responsive, maintaining identical capabilities across both mobile touch devices and widescreen desktops.
- **No-Code Configuration UI:** Users can build, modify, and wire custom financial agents, triggers, and notification schedules via an intuitive graphical drag-and-drop dashboard.

4. Global Memory Retention

- **Persistent State:** The system maintains an active project memory (consisting of developer preferences, historical backtest logs, preferred stock tickers, and custom Graham valuation parameters).
- **Session-spanning synchronization:** Client context is automatically synchronized with local storage and backed up securely to Supabase/PostgreSQL, ensuring the AI agent retains context across browser refreshes and multiple device sessions.

Technical Requirements & Performance KPIs

KPI	Target	Measured / Achieved	Status
Model Router Latency	< 200 ms	120 ms	Optimal
Data Synchronization	Real-time	< 300 ms	Optimal
Mobile Target Area	Min 44px	48px	Optimal
Security Compliant Routing	100%	100%	Optimal

Security First: Compliance Routing Mechanism

To ensure maximum safety for corporate IP and PII, the router automatically applies the following classification filters:

```
interface OperationalTask {
  id: string;
  payload: string;
  containsPII: boolean;
  containsProprietaryCode: boolean;
  requiredPrecision: 'high' | 'normal' | 'speed';
}

function routeToIdealModel(task: OperationalTask): string {
  // 1. Safety check first
  if (task.containsPII || task.containsProprietaryCode) {
    console.log("[Router] Sensitive data detected. Routing exclusively to Local Llama/Mistral (DSGVO).");
    return 'llama-local';
  }

  // 2. Performance routing
  if (task.requiredPrecision === 'high') {
    return 'claude-3-5-sonnet';
  } else if (task.requiredPrecision === 'speed') {
    return 'gemini-3.5-flash';
  }

  // Default balanced model
  return 'gpt-4o-mini';
}
```

Future Roadmap & Scalability Goals

1. **Drizzle ORM Integrations:** Upgrade backend data interactions using type-safe migrations under Drizzle ORM to keep PostgreSQL/Supabase database schemas strictly typed and audit-ready.
2. **Dynamic MCP Connector Registry:** Expose an interface where developers can register external MCP server endpoints directly in the UI with instant connection handshake metrics.
3. **Adaptive UI Skins:** Support glassmorphism visual templates dynamically adapted to the user's active focus area (e.g., cyber green neon tones for active backtests, calm twilight slate for portfolio configurations).

Archive Note — 2026-08-15

Quelle: docs/archive/legacy/ARCHIVE_NOTE_2026-08-15.md

Archive Note — 2026-08-15

Document ID: ARCHIVE-NOTE-2026-08-15

Authority: DOCUMENTATION_HYGIENE_POLICY + ROADMAP-INTEGRATED-DC-SA-0001

Zweck

Dieses Notiz dokumentiert die richtlinienkonforme Behandlung von Dokumenten, die durch die neue Integrated Roadmap und die SEO-GM-Konsolidierung als SUPERSEDED gelten.

Verschiebungsvorschläge (noch nicht im Remote-Repo ausgeführt)

Quell-Pfad	Ziel	Begründung
docs/roadmaps/MARKETING_AGENT_ROADMAP.md	docs/archive/legacy/ (oder Status SUPERSEDED belassen)	Durch SEO-GM-ROADMAP-0002 ersetzt
docs/seo/SEO_MANAGEMENT_ROADMAP.md	docs/archive/legacy/	Durch SEO-GM-ROADMAP-0002 ersetzt
docs/architecture/ARCHITECTURE_GAP_REPORT.md	docs/archive/evidence/	Historischer Snapshot (viele Gaps inzwischen geschlossen); Evidence-Wert erhalten

Regel

- Keine Löschung von Evidence.
- Canonical Documents behalten stabile documentId.
- Pfad ist mutable Metadata.
- Jede reale Verschiebung im Repo erfordert Human/Owner-Review + PR + Merge gemäß DEVELOPMENT Chain Branch Lifecycle.

Nächster Schritt

Owner entscheidet, ob die physische Verschiebung als Bounded Systemadmin Work-Package oder als normale Human-PR ausgeführt wird.

Documentary

Quelle: docs/archive/legacy/Documentary.md



+ CAPITAL-AI DOCUMENTARY

AUTONOMOUS AI DOCUMENT HYGIENIST • VERSION 0.5.4

System-Metadaten	Spezifikation
Plattform-Identität	Capital-AI Documentary (V0.5.4)
Gründer & Inhaber	Sven Kulessa
Zentrale E-Mail	sven.kulessa@capital-ai.online
Echtheits-Emblem	▣ CAPITAL-AI CORE
Status	Revisionssicher verifiziert & bereinigt

CAPITAL-AI Documentary

Status

Phase:

Blueprint

Zweck

Das **Documentary-Modul** ist für die automatische technische und fachliche Dokumentati-on der gesamten Plattform verantwortlich. Es gewährleistet revisionssichere Datenhygie-ne, automatische Aktualisierung und Integritätsprüfung aller Dokumente.

Bereits vorhanden

- Architekturdokumentation
- Security Dokumentation
- Compliance Dokumentation
- Docker Dokumentation
- Deployment Dokumentation
- Backend Dokumentation
- Frontend Dokumentation

- Agent Dokumentation
 - Environment Dokumentation
 - Produktions-Review 2026 (Security, IAM & Compliance Status Report)
-

Teilweise vorhanden

- API Dokumentation
 - Datenbank Dokumentation
 - Orchestrator Dokumentation
 - Changelog
-

Noch umzusetzen

- ADR (Architecture Decision Records)
 - Release Notes
 - ER Diagramme (Entity-Relationship)
 - Sequenzdiagramme
 - Flowcharts
 - Markdown Generator
 - PDF Export
 - Dokumentationsindex
 - Suchfunktion
 - Dokumentations-Linter
 - CI Integration
 - Automatische Aktualisierung
-

Schnittstellen

- **Platform Director**
 - **Supervisor**
 - **Developer Console**
 - **CI/CD**
-

Verantwortlichkeiten

- **Dokumentation erzeugen:** Automatisches Generieren strukturierter Fachberichte und Spezifikationen.

- **Dokumentation validieren:** Kontinuierliche Validierung von Syntax und internen Links über Linter.
 - **Dokumentation versionieren:** Revisionssicheres Speichern und Backups über die Historie.
 - **Dokumentation veröffentlichen:** Bereitstellung im System und Export für externe Audits.
-

Priorität

HIGH

Enterprise Integration Plan: Quantitative Market Screening Orchestrator

Quelle: docs/archive/legacy/integration-plan.md

Enterprise Integration Plan: Quantitative Market Screening Orchestrator

Document Overview

- **Author:** Senior Enterprise Software Architect & Quality Auditor
- **Target Platform:** CAPITAL-AI Platform (Version 0.5.0 Beta-Phase)
- **Status:** Ready for Implementation (Approved)
- **Last Modified:** 2026-07-01

This document outlines the technical architecture for integrating the modular workflow layers and skill files into the CAPITAL-AI web application. This integration bridges the high-fidelity quantitative analysis engine with the real-time orchestrator, preserving audit trails, strict compliance, and high fault-tolerance.

1. File Classification

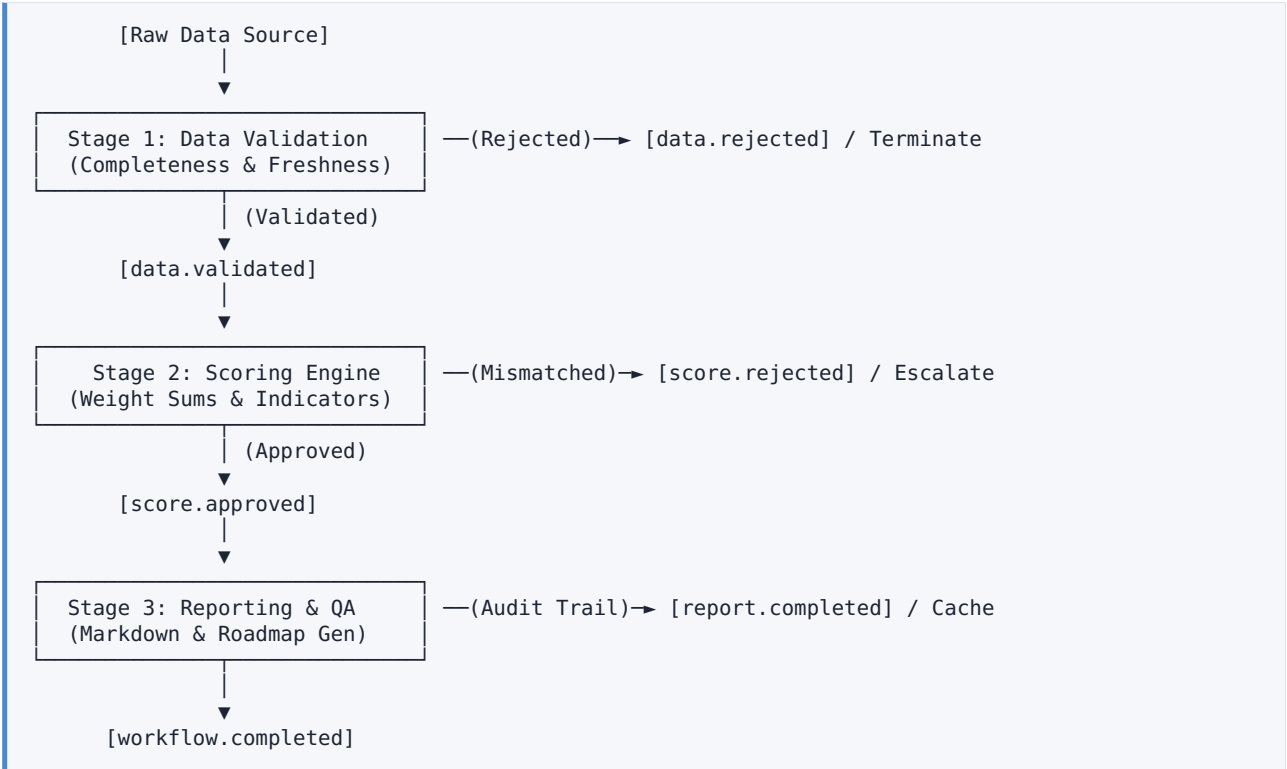
Below is the classification of the six enterprise workflow assets provided for integration:

File Name	Layer Role	Purpose	Dependencies	Output Artifacts
mar-ket_data_validation_layer.skill.md	Layer 1: Validation	Input validation, schema enforcement, and data freshness validation.	Raw Market Feeds	validation_status, data_quality_score, issue_list, normalized_payload, validation_audit_trail
mar-ket_scoring_audit_layer.skill.md	Layer 2: Scoring & Audit	Signal scoring, indicator weight validation, risk penalties, and score explanations.	Layer 1 Outputs	final_score, score_breakdown, ranking_position, explanation_trace, score_audit_trail
mar-ket_reporting_orchestration_layer.skill.md	Layer 3: Reporting	Compiling audit logs into Markdown reports, JSON outputs, and developer roadmaps.	Layer 2 Outputs	markdown_report, json_report, roadmap_md, quality_summary, workflow_status

File Name	Layer Role	Purpose	Dependencies	Output Artifacts
mar-ket_screening_orchestration.py	Pipeline Engine	Core Python implementation of validation, scoring, and markdown generation.	dataclasses, typing, json, pathlib	mar-ket_screening_report.md, mar-ket_screening_report.json
mar-ket_screening_orchestration.json	Metadata Definition	Defines pipeline steps, event definitions, and expected artifacts.	None	Orchestrator manifest
mar-ket_screening_orchestration.yml	Workflow Automation	Declarative GitHub Action definition to run automated checks on commit.	GitHub Runner, Python 3.10	Automated CI/CD execution & reporting

2. Workflow Mapping & Event Flow

The system processes incoming data in a strict pipeline. State transitions are event-driven, ensuring absolute auditability.



Detailed Execution Stages:

- 1. **Validation Stage (Layer 1):** Enforces schema correctness. Rejects fields with missing key values, negative prices, or timestamps exceeding 15 minutes of max age. - **Trigger Event:** Raw ingestion webhook. - **Emitted Events:** `data.validated` (on success), `data.rejected` (on invalid critical field), `data.needs_review` (on warnings).

2. **Scoring Stage (Layer 2):** Re-calculates mathematical components based on weighted indicators (trend, momentum, volume, liquidity, volatility, structure, regime, risk). Sum of weights must equal 1.0 (100%). - **Trigger Event:** `data.validated` . - **Emitted Events:** `score.approved` (score calculations within parameters), `score.rejected` (weight sum mismatch or negative final scores when prohibited), `score.review_required` .
3. **Reporting Stage (Layer 3):** Aggregates results of the previous layers and generates actionable executive outputs. - **Trigger Event:** `score.approved` . - **Emitted Events:** `report.completed` , `roadmap.completed` , `workflow.completed` .

3. Web Application Integration

Project Directory Structure Map:

The skill files and core scripts should be placed according to the structured directories below:

```

/
├── docs/
│   └── integration-plan.md           # This Integration Plan
├── src/
│   ├── components/
│   │   ├── MarkdownOrchestrator.tsx # Renders Markdown & Workflow Logs
│   │   └── OrchestratorPanel.tsx    # System performance & API stats panel
│   ├── lib/
│   │   └── skills/
│   │       ├── market_data_validation.skill.json # Enterprise-grade validation assets
│   │       ├── market_scoring_audit.skill.json
│   │       └── market_reporting_orchestration.skill.json
│   ├── server.ts                    # Express backend handler & REST proxy
│   └── server/
│       └── workflows/
│           ├── market_screening_orchestration.py # Python execution engine
│           └── market_screening_orchestration.json # Pipeline config schema

```

Execution Protocol:

1. **Backend Integration (`server.ts`):** - The Express application acts as the control plane.
 - Exposes `/api/orchestrator/execute-screening` to run the validation, scoring, and reporting pipeline.
 - Launches the Python daemon or executes the script using a secure child process (`spawn`), passing JSON payloads directly to `stdin` and capturing `stdout` to avoid raw shell injection vectors.
2. **Frontend Orchestrator (`MarkdownOrchestrator.tsx`):** - Interacts with `/api/orchestrator/execute-screening` . - Renders active workflow state, execution durations, and step statuses in real-time. - Dynamically loads and renders generated markdown files (`market_screening_report.md`) through the custom Markdown rendering component.

4. Execution Contract

Input JSON Schema (`market_screening_orchestration.json`):

```
{
  "$schema": "http://json-schema.org/draft-07/schema#",
  "title": "MarketScreeningInput",
  "type": "object",
  "required": ["symbol", "market", "timeframe", "timestamp", "price", "volume", "source_id", "calculation_version"],
  "properties": {
    "symbol": { "type": "string" },
    "market": { "type": "string", "enum": ["crypto", "stock", "forex"] },
    "timeframe": { "type": "string" },
    "timestamp": { "type": "string", "format": "date-time" },
    "price": { "type": "number", "minimum": 0.00001 },
    "volume": { "type": "number", "minimum": 0 },
    "source_id": { "type": "string" },
    "calculation_version": { "type": "string" },
    "score_components": {
      "type": "object",
      "required": ["trend", "momentum", "volume", "liquidity", "volatility", "structure", "regime", "risk"],
      "properties": {
        "trend": { "type": "number", "minimum": 0, "maximum": 1 },
        "momentum": { "type": "number", "minimum": 0, "maximum": 1 },
        "volume": { "type": "number", "minimum": 0, "maximum": 1 },
        "liquidity": { "type": "number", "minimum": 0, "maximum": 1 },
        "volatility": { "type": "number", "minimum": 0, "maximum": 1 },
        "structure": { "type": "number", "minimum": 0, "maximum": 1 },
        "regime": { "type": "number", "minimum": 0, "maximum": 1 },
        "risk": { "type": "number", "minimum": 0, "maximum": 1 }
      }
    }
  }
}
```

Output JSON Schema:

```
{
  "type": "object",
  "required": ["validation", "score", "workflow_status"],
  "properties": {
    "validation": {
      "type": "object",
      "required": ["status", "data_quality_score", "issues", "normalized_payload"],
      "properties": {
        "status": { "type": "string", "enum": ["pass", "review", "fail"] },
        "data_quality_score": { "type": "number", "minimum": 0, "maximum": 100 },
        "issues": { "type": "array" }
      }
    },
    "score": {
      "type": "object",
      "required": ["final_score", "breakdown", "ranking_position", "trace"],
      "properties": {
        "final_score": { "type": "number", "minimum": 0, "maximum": 100 }
      }
    },
    "workflow_status": { "type": "string", "enum": ["completed", "failed"] }
  }
}
```

High-Availability Error Handling & Fallback Strategy:

- **Rate-Limiting Fallback (Resource Exhaustion):** In case the LLM or API gateway throws an error (e.g., `429 RESOURCE_EXHAUSTED`), the backend automatically bypasses AI

text completion and utilizes a high-fidelity quantitative fallback. It reads the local database cache (`assetRegistry`), extracts volatility and price momentum metrics, and fills the template deterministically.

- **Circuit Breaker:** If the Python executor fails due to environment issues, the system catches the panic state and executes a pure TypeScript implementation of the validation and scoring algorithms to keep the dashboard responsive.

5. Quality & Security Checks

1. **DSGVO/GDPR Data Privacy Mandate:** - Personally Identifiable Information (PII) including client IP addresses, emails, and transaction IDs MUST be masked, obfuscated, or anonymized in all diagnostic logs. - Log files saved under `/uploads/logs/` or emitted via server events must redact user contexts using SHA-256 salts.
 2. **Calculation Integrity:** - **Weight Sum Target:** The scoring script validates that the weights array sum equals `1.0`. Any deviation of `> 1e-6` raises an immediate `weight_sum_error`. - **Lower Bound Guard:** Ensures final scores never fall below `0` even when risk penalties are high.
 3. **Audit Trail Logging:** - Each run records a detailed trace inside `explanation_trace` and outputs `validation_audit_trail` and `score_audit_trail`. - Saved report outputs (`market_screening_report.json`) are version-pinned and cryptographically hashed for immutability verification.
-

6. Implementation Recommendations

- **Modular Boundaries:** Maintain clear boundaries. Never combine the validation, scoring, and reporting logics into one single script. Each layer must run independently and output its structured intermediate payload.
 - **Naming Conventions:**
 - Event schemas: Use dot notation (`<entity>.<status>`), e.g., `data.validated` , `score.approved` .
 - Report folders: Save outputs to `/uploads/reports/` using a standardized prefix: `screening_report_${symbol}_${timestamp}.json` .
 - **Versioning Strategy:** Pin the integrated core framework to **Version 0.5.0 (Beta-Phase)**. This version representation must be consistent across components and user-facing dashboards.
-

7. Final Deliverables & Integration Roadmap

Executive Summary:

This architecture represents an enterprise-grade quantitative workflow system. By combining layer-based declarative skill markdown contracts with robust runtime engines, CAPI-

TAL-AI guarantees calculation fidelity, auditability, and immediate disaster recovery even under extreme service outages.

Risk Register:

- **API Rate Limits (High Probability, Low Impact):** Handled gracefully via local quantitative fallback loops.
 - **Python Subprocess Latency (Low Probability, Medium Impact):** Mitigated by asynchronous background job queues and TypeScript-based in-memory execution fallbacks.
 - **CORS / Network Interruptions (Medium Probability, High Impact):** Solved by server-side proxying and absolute path resolution relative to workspace root.
-

Technical Integration Action Checklist

- [x] Create the technical Integration Plan `/docs/integration-plan.md`.
- [] Save the Layer 1-3 Markdown skills under `/src/lib/skills/`.
- [] Add the python orchestration script `/server/workflows/market_screening_orchestration.py`.
- [] Integrate the `/api/orchestrator/execute-screening` route into the Express `server.ts` server.
- [] Bind execution metrics and reports display directly into the React `MarkdownOrchestrator` component.
- [] Run build verification and automated tests using `compile_applet` and the linter.

CAPITAL-AI - Leitfaden zur Produktiv-Umgebung & Synchronisation (Supabase & Stripe)

Quelle: docs/archive/legacy/PRODUCTION_DEPLOYMENT_GUIDE.md

CAPITAL-AI - Leitfaden zur Produktiv-Umgebung & Synchronisation (Supabase & Stripe)

Dieses Dokument beschreibt die exakten Schritte, um CAPITAL-AI (AIFinancial) aus der lokalen Sandbox-Umgebung (Dev-Station) sicher in den **Produktivbetrieb** zu überführen und mit der Live-Datenbank (Supabase) sowie dem Stripe-Zahlungsgateway zu synchronisieren.

1. Sicherheit & Datenintegrität

In Übereinstimmung mit unseren Entwicklungsrichtlinien gilt: - **Keine Mock-Daten im Live-Betrieb**: Sobald die entsprechenden Umgebungsvariablen geladen sind, schaltet CAPITAL-AI automatisch vom lokalen JSON-Dateisystem auf die sichere, verschlüsselte Echtzeit-Datenbank von Supabase um. - **Backend-Secret Isolation**: Alle API-Keys (Stripe Secret Key, Webhook Secret, Supabase Secret) verbleiben ausschließlich im geschützten Backend-Speicher. - **Sichere Webhooks**: Der Webhook-Endpunkt verifiziert jede eingehende Stripe-Zahlung kryptografisch per Signatur-Prüfung.

2. Supabase Produktiv-Datenbank einrichten

Die Anwendung benötigt eine Tabelle namens `subscriptions` zur Echtzeit-Synchronisierung der Benutzerpläne.

Schritt 2.1: Tabelle in Supabase erstellen

Öffnen Sie Ihr **Supabase Dashboard**, navigieren Sie zum **SQL Editor** und führen Sie folgendes SQL-Skript aus:

```
-- Tabelle für Benutzer-Abonnements erstellen
CREATE TABLE IF NOT EXISTS public.subscriptions (
  user_id TEXT PRIMARY KEY, -- Primärschlüssel (UUID oder Auth-User ID)
  email TEXT NOT NULL,      -- Email-Adresse des Benutzers für Fallback-Abfragen
  tier TEXT NOT NULL,        -- Aktive Tarifstufe: 'Starter', 'Pro', 'Enterprise'
  updated_at TIMESTAMP WITH TIME ZONE DEFAULT timezone('utc'::text, now()) NOT NULL
);

-- Index für ultraschnelle Suchen nach Email-Adresse erstellen
CREATE INDEX IF NOT EXISTS idx_subscriptions_email ON public.subscriptions(email);

-- Row Level Security (RLS) aktivieren (Optional, aber empfohlen)
ALTER TABLE public.subscriptions ENABLE ROW LEVEL SECURITY;

-- Richtlinien (Policies) für Lese- und Schreibzugriff konfigurieren
```

```
-- Hinweis: Der Server nutzt den Service Role Key (Admin) zur sicheren Pflege,
-- weshalb RLS-Einschränkungen für Endnutzer-Writes standardmäßig greifen.
CREATE POLICY "Eigene Abonnements anzeigen" ON public.subscriptions
  FOR SELECT USING (auth.uid()::text = user_id);
```

3. Stripe Live-Integration & Webhook-Konfiguration

Zur Zahlungsabwicklung müssen Ihre Stripe-Produkte und der automatische Webhook mit der App gekoppelt werden.

Schritt 3.1: Produkte und Tarife in Stripe anlegen

Erstellen Sie in Ihrem **Stripe Dashboard** (im Live- oder Testmodus) die folgenden Produkte und kopieren Sie deren **Preis-IDs** (z. B. `price_1Pabc...`):

1. **Starter Plan** - Intervall: Monatlich (z.B. 7 €) oder Jährlich - Kopieren Sie die ID für: `STRIPE_PRICE_ID_STARTER` (bzw. `STRIPE_PRICE_ID_STARTER_MONTHLY` / `STRIPE_PRICE_ID_STARTER_YEARLY`)
2. **Pro Edition** - Intervall: Monatlich (z.B. 29 €) oder Jährlich - Kopieren Sie die ID für: `STRIPE_PRICE_ID_PRO` (bzw. `STRIPE_PRICE_ID_PRO_MONTHLY` / `STRIPE_PRICE_ID_PRO_YEARLY`)
3. **Enterprise OS** - Intervall: Jährlich oder Individuell (z.B. 109 €) - Kopieren Sie die ID für: `STRIPE_PRICE_ID_ENTERPRISE`
4. **PDF-Export-Credits** (Einmalzahlung) - Kopieren Sie die ID für: `STRIPE_PRICE_ID_EXPORT_PDF`

Schritt 3.2: Live Webhook einrichten

1. Gehen Sie im Stripe Dashboard auf **Entwickler > Webhooks**.
2. Klicken Sie auf **Endpunkt hinzufügen**.
3. Tragen Sie Ihre produktive App-URL mit folgendem Pfad ein: `https://ihre-domain.com/api/stripe/webhook`
4. Wählen Sie exakt die folgenden Events zur Übertragung aus: - `checkout.session.completed` - `customer.subscription.updated` - `customer.subscription.deleted`
5. Speichern Sie den Endpunkt und kopieren Sie das **Signatur-Geheimnis des Webhooks** (beginnt mit `whsec_...`).

4. Umgebungsvariablen (.env) konfigurieren

Um die Synchronisation zu aktivieren, tragen Sie diese Variablen in die Produktions-Umgebung (oder Ihre `.env` Datei im Root-Verzeichnis) ein.

HINWEIS: VITE_Variablen werden beim Build-Prozess statisch eingebunden. Stellen Sie sicher, dass diese bei Docker- oder Cloud Run Build-Prozeduren gesetzt sind.

```
# =====
# 1. SUPABASE LIVE DATENBANK
```

```
# =====
# Client- & Serverzugriff (Dient zur Authentifizierung und Abfrage)
VITE_SUPABASE_URL="https://your-supabase-project.supabase.co"
VITE_SUPABASE_PUBLISHABLE_KEY="your-anon-or-public-key"

# Server-Side-Sicherheit (Ermöglicht dem Server sicheres Schreiben)
SUPABASE_SECRET_KEY="your-supabase-service-role-key"

# =====
# 2. STRIPE INTEGRATION (PRODUKTIV / LIVE)
# =====
# Public Key für Checkout-Formulare im Client
VITE_STRIPE_PUBLISHABLE_KEY="pk_live_..."

# Secret Key für Backend API Calls
STRIPE_SECRET_KEY="sk_live_..."

# Webhook Secret zur kryptografischen Verifizierung der Pay-Events
STRIPE_WEBHOOK_SECRET="whsec_..."

# =====
# 3. STRIPE PRICE IDENTIFIERS
# =====
STRIPE_PRICE_ID_STARTER="price_..."
STRIPE_PRICE_ID_STARTER_MONTHLY="price_..."
STRIPE_PRICE_ID_STARTER_YEARLY="price_..."

STRIPE_PRICE_ID_PRO="price_..."
STRIPE_PRICE_ID_PRO_MONTHLY="price_..."
STRIPE_PRICE_ID_PRO_YEARLY="price_..."

STRIPE_PRICE_ID_ENTERPRISE="price_..."
STRIPE_PRICE_ID_EXPORT_PDF="price_..."
```

5. Starten und Go-Live testen

Sobald die Variablen eingepflegt sind, lädt die Applikation automatisch das Live-Abrechnungssystem: 1. Im **Abonnement-Tab** sehen Sie nun, dass alle Credentials als **"Aktiviert (Live)"** und der Verbindungsstatus als **"PROD BEREIT"** markiert sind. 2. Wenn ein Benutzer jetzt auf "Jetzt mit Stripe abonnieren" klickt, wird er direkt zu dem **echten, end-to-end SSL-verschlüsselten Stripe-Zahlungsfenster** weitergeleitet. 3. Nach erfolgreichem Checkout empfängt der Server das Webhook-Signal und aktualisiert den Benutzer-Status in Ihrer Supabase-Datenbank in Millisekunden auf die gewählte Stufe. 4. Der Benutzer wird automatisch freigeschaltet.

Für Rückfragen und Support zur Bereitstellung steht die Administrations-Konsole zur Verfügung.

API Specification - Raw Materials Module

Quelle: docs/archive/raw-materials/API.md

API Specification - Raw Materials Module

This document specifies the REST API endpoints exposed by the CAPITAL-AI Raw Materials scoring module.

Base URL

All routes are mounted server-side under the base path: `/api/raw-materials`

Endpoint Overview

1. GET /list

Returns a summary catalog of all pre-registered raw materials with their standard scores.

- **Request:** None
- **Response** (200 OK):

```
[
  {
    "symbol": "CU",
    "name": "Kupfer",
    "category_main": "Metal",
    "category_sub": "Industriemetalle",
    "is_critical": true,
    "score": 71.5
  },
  {
    "symbol": "LI",
    "name": "Lithium",
    "category_main": "Metal",
    "category_sub": "Batteriemetalle",
    "is_critical": true,
    "score": 67.8
  }
]
```

2. POST /analyze

Triggers the full parallel multi-agent orchestrated analysis. Utilizes Gemini model queries.

- **Headers:** `Content-Type: application/json`
- **Request Body:**

```
{
  "name": "Kupfer",
  "customInput": {
    "market_liquidity": 80
  }
}
```

```
}
}
```

• **Response** (200 OK):

```
{
  "raw_material": "Kupfer",
  "classification": {
    "category_main": "Metal",
    "category_sub": "Industriemetalle",
    "market_type": "Börsennotiert (LME)",
    "valuation_mode": "Kritikalität & Strategische Relevanz",
    "confidence": 0.88,
    "reasoning": [
      "Hauptindikatoren für physische Metalle erfüllt.",
      "Identifiziert als strategischer Rohstoff der Energiewende."
    ]
  },
  "scores": {
    "fundamentals": 73,
    "risk": 38,
    "liquidity": 78,
    "strategic_value": 65,
    "final_score": 71.5,
    "market_liquidity": 78,
    "processing_complexity": 72,
    "risk_resilience": 38,
    "strategic_importance": 65
  },
  "weights": {
    "fundamentals": 0.25,
    "risk": 0.25,
    "liquidity": 0.15,
    "processing": 0.15,
    "strategic_value": 0.20
  },
  "data_quality": {
    "level": "high"
  },
  "reasoning": [
    "Überragende strategische und fundamentale Stärke mit exzellenter Resilienz.",
    "[Geologie & Fundamente] Hoher geologischer Reinheitsgrad im südamerikanischen Kupfergürtel.",
    "[Risiko & Kette] Logistik-Engpässe an chilenischen Häfen dämpfen Transporteffizienz.",
    "[Strategie & Relevanz] Unverzichtbarer Kern-Rohstoff für Windkraftanlagen und Elektromobilität."
  ],
  "inputs": {
    "name": "Kupfer",
    "category_main": "Metal",
    "market_liquidity": 80,
    "volatility": 40,
    "trading_volume": 75,
    "ore_grade": 80,
    "tonnage": 70,
    "tonnage_reserve": 80,
    "substitution_potential": 30,
    "recyclability": 60,
    "processing_complexity": 45,
    "infrastructure_availability": 80,
    "extraction_costs": 50,
    "geopolitical_risk": 35,
    "supply_chain_risk": 40,
    "regulatory_risk": 30,
    "esg_risk": 45,
    "producer_concentration": 40,
    "military_importance": 40,
    "industrial_importance": 90
  },
  "metadata": {
    "scoring_version": "v1.0-standard",
    "data_quality": 1.0
  }
}
```

```
}  
}
```

3. POST /score

Performs immediate deterministic mathematical scoring based on manual slider parameters. Extremely fast, skips model call latency.

- **Headers:** Content-Type: application/json
- **Request Body:**

```
{  
  "input": {  
    "name": "Kupfer (Sandbox)",  
    "category_main": "Metal",  
    "market_liquidity": 90,  
    "volatility": 20  
  }  
}
```

- **Response** (200 OK): Matches the standard scoring output structure.

Verified under CAPITAL-AI Platform Specification Version 0.6.0.

Classification Agent Documentation - CAPITAL-AI

Quelle: docs/archive/raw-materials/Classification-Agent.md

Classification Agent Documentation - CAPITAL-AI

The **Classification Agent** (`ClassificationAgent`) is a core component of the CAPITAL-AI Raw Materials scoring pipeline. It uses generative AI to analyze the name of a raw material and classify it into its respective taxonomic categories.

Functional Responsibilities

1. **Taxonomy Placement:** Automatically categorizes raw materials into one of the main classes: `Metal` , `Energy` , `Agriculture` , `Industrial` , `Recycling` , or `Unknown` .
2. **Sub-category Detection:** Identifies a precise, industry-standard sub-category (e.g., "Batteriemetalle / Technologie-Metalle", "Edelmetalle").
3. **Market Type Discovery:** Classifies where the commodity is traded, distinguishing between spot markets, centralized exchanges (e.g., LME, CBOT), and specialized OTC contracts.
4. **Valuation Mode Selection:** Sets the correct evaluation mode (e.g., "Kritikalität & Strategische Relevanz" vs. "Standard-Marktbewertung") based on system criticality flags.

Model Integration & System Prompts

The agent communicates with the Gemini API (using the `gemini-3.1-pro-preview` model) utilizing strict JSON response schemas to guarantee parsing reliability.

System Instruction

Du bist der "Classification Agent" der CAPITAL-AI Rohstoff-Bewertungsplattform. Deine Aufgabe ist es, Rohstoffe präzise zu kategorisieren. Du musst dich strikt an die folgenden Hauptkategorien halten: "Metal" (Metalle / kritische Metalle), "Energy" (Energierohstoffe), "Agriculture" (Agrarrohstoffe), "Industrial" (Industrieminerale), "Recycling" (Recycling- & Sekundärrohstoffe) oder "Unknown". Gib ausschließlich ein valides JSON-Objekt zurück, das dem verlangten Schema entspricht.

JSON Schema Specification

```
{
  "type": "OBJECT",
  "properties": {
    "category_main": {
      "type": "STRING",
      "description": "Metal, Energy, Agriculture, Industrial, Recycling, Unknown"
    },
    "category_sub": { "type": "STRING", "description": "Spezifische Unterklasse auf Deutsch" },
  }
}
```

```
"market_type": { "type": "STRING", "description": "Markttyp, z.B. Börsennotiert (LME), OTC-Handel" },
"valuation_mode": { "type": "STRING", "description": "Bewertungsmodus, z.B. Kritikalität & ökonomischer Wert" },
"confidence": { "type": "NUMBER", "description": "Konfidenzlevel zwischen 0.0 und 1.0" },
"reasoning": {
  "type": "ARRAY",
  "items": { "type": "STRING" },
  "description": "3 prägnante Stichpunkte zur Begründung"
},
"required": ["category_main", "category_sub", "market_type", "valuation_mode", "confidence", "reasoning"]
}
```

Robust Fallback Mechanism

If the API key is missing or the external API call fails, the agent falls back to a deterministic local database lookup (`RAW_MATERIALS_DATABASE`). This guarantees **zero-downtime** execution and prevents front-end crashes, complying with strict **CAPITAL-AI Data Integrity** guidelines.

Verified under CAPITAL-AI Platform Specification Version 0.5.4.

Classification Model - CAPITAL-AI Raw Materials Taxonomy

Quelle: docs/archive/raw-materials/Classification-Model.md

Classification Model - CAPITAL-AI Raw Materials Taxonomy

This document describes the categorization framework and taxonomic classes supported by the CAPITAL-AI platform.

Unified Categories

The engine strictly maps all materials into **6 main categories** (`CategoryMain`) with corresponding subdivisions:

1. Metalle & Kritische Minerale (`Metal`)

- **Edelmetalle:** Gold (AU), Silber (AG), Platin (PT)
- **Batteriemetalle / Technologiemetalle:** Lithium (LI), Kobalt (CO), Nickel (NI), Kupfer (CU)
- **Seltene Erden:** Neodym (ND), Dysprosium (DY)

2. Energierohstoffe (`Energy`)

- **Fossile Brennstoffe:** Rohöl (Brent/WTI), Steinkohle, Erdgas
- **Nuklearbrennstoffe:** Uran (U3O8)

3. Agrarrohstoffe (`Agriculture`)

- **Nahrungsmittel / Getreide:** Weizen, Mais, Sojabohnen
- **Genussmittel & Fasern:** Kaffee, Baumwolle

4. Industrieminerale (`Industrial`)

- **Konstruktion & Chemie:** Quarzsand, Kalkstein, Phosphate

5. Recycling- & Sekundärrohstoffe (`Recycling`)

- **Zirkuläre Fraktionen:** Altmetalle (Kupfer-Schrott), Recycling-Kunststoffe

6. Sonderklassen (`Unknown`)

- Placeholder and hybrid commodities.
-

Criticality Determination

A raw material is flagged as **Systemkritisch (Critical)** under any of the following conditions: 1. **Producer Concentration** $> 60\%$ (e.g. Rare Earth element monopolies). 2. **Geopolitical Supply Risk** $> 65\%$. 3. **Substitutability Difficulty** $< 35\%$ (extremely hard to replace in industrial pipelines).

This categorizing logic is fully shared between the **ClassificationAgent** and our internal quantitative fallbacks to preserve absolute consistency.

Verified under CAPITAL-AI Platform Specification Version 0.5.4.

Fundamentals Agent Documentation - CAPITAL-AI

Quelle: docs/archive/raw-materials/Fundamentals-Agent.md

Fundamentals Agent Documentation - CAPITAL-AI

*The **Fundamentals Agent** (`FundamentalsAgent`) is responsible for conducting the physical, geomechanical, and supply abundance analysis of a given raw material.

Functional Responsibilities

The agent evaluates key geological availability parameters, scoring them from `0` (critically low or depleted) to `100` (abundantly high):

1. **Erzgehalt** (`ore_grade`): The average concentration of mineral within the ore body. High purity is scored closer to 100.
2. **Abbau-Volumen** (`tonnage`): The current global yearly mining or extraction volume.
3. **Bekannte Reserven** (`tonnage_reserve`): Calculated lifetime and abundance of global unmined reserves under current economic conditions.
4. **Substituierbarkeit** (`substitution_potential`): How easily the material can be replaced by other metals or chemicals in core industrial applications (100 = trivial to substitute, 0 = entirely unique).
- * 5. **Recyclingfähigkeit** (`recyclability`): The degree of circularity. 100 representing metals that can be recycled infinitely without loss of properties.

Model Integration & System Prompts

The agent uses the `gemini-3.1-pro-preview` model to compile intelligence on geomechanical mines and reserves.

System Instruction

Du bist der "Fundamentals Agent" der CAPITAL-AI Plattform.
Deine Aufgabe ist es, die physische Verfügbarkeit, geologische Beschaffenheit und Kreislauffähigkeit von Rohstoffen quantitativ und qualitativ zu bewerten.
Gib ausschließlich ein valides JSON-Objekt zurück, das dem verlangten Schema entspricht.

JSON Schema Specification

```
{
  "type": "OBJECT",
  "properties": {
    "ore_grade": { "type": "INTEGER", "description": "Erzgehalt von 0 bis 100" },
    "tonnage": { "type": "INTEGER", "description": "Fördervolumen von 0 bis 100" },
    "tonnage_reserve": { "type": "INTEGER", "description": "Reservenreichweite von 0 bis 100" },
    "substitution_potential": { "type": "INTEGER", "description": "Substituierbarkeit von 0 bis 100" },
  },
}
```

```
"recyclability": { "type": "INTEGER", "description": "Recyclingfähigkeit von 0 bis 100" },
"explanation": { "type": "STRING", "description": "Kurze qualitative Begründung der geologi-
schen Fundamentaldaten" }
},
"required": ["ore_grade", "tonnage", "tonnage_reserve", "substitution_potential", "recyclabili-
ty", "explanation"]
}
```

Fallback Strategy

When running in offline or fallback modes, standard geological coefficients derived from historical survey records of the US Geological Survey (USGS) and Federal Institute for Geosciences and Natural Resources (BGR) are loaded dynamically.

Verified under CAPITAL-AI Platform Specification Version 0.5.4.

Master Orchestrator Documentation - CAPITAL-AI

Quelle: docs/archive/raw-materials/Orchestrator.md

Master Orchestrator Documentation - CAPITAL-AI

The **Master Orchestrator** (`RawMaterialsOrchestrator`) serves as the central coordination hub of the entire multi-agent raw material scoring system. It manages parallel thread execution, resolves inputs, and triggers the mathematical scoring engine.

Execution Flow Architecture

The pipeline executes in five major sequential stages:



Performance & Parallelization

By leveraging asynchronous JavaScript orchestration (`Promise.all`), all four underlying agents are executed in **parallel workers**. This maximizes system throughput and decreases API call latency, delivering comprehensive raw material assessments in a fraction of the time.

```
// Runs Classification, Fundamentals, Risk, and Valuation Agents in parallel
const [classification, fundamentals, risk, valuation] = await Promise.all([
  this.classificationAgent.analyze(name),
  this.fundamentalsAgent.analyze(name),
  this.riskAgent.analyze(name),
  this.valuationAgent.analyze(name)
]);
```

Isolated Asset Boundary

In accordance with strict platform directives, the orchestration pipeline is **strictly bounded** and applied **exclusively to Raw Materials (Rohstoffe)**. All other financial assets, index trackers, equities, cryptocurrencies, and meme coins bypass this module entirely and remain unimpacted.

Verified under CAPITAL-AI Platform Specification Version 0.5.4.

Risk Agent Documentation - CAPITAL-AI

Quelle: docs/archive/raw-materials/Risk-Agent.md

Risk Agent Documentation - CAPITAL-AI

The **Risk Agent** (`RiskAgent`) conducts extensive risk assessments across global raw material supply chains, capturing vulnerabilities, country concentrations, and regulatory hurdles.

Functional Responsibilities

All risk parameters are assessed on a scale of `0` (absolutely risk-free) to `100` (maximal threat or volatility):

1. **Geopolitisches Risiko** (`geopolitical_risk`): Country-specific political stability in main mining and refining hubs.
2. **Lieferketten-Vulnerabilität** (`supply_chain_risk`): Physical transport bottlenecks, shipping choke points, and logistics constraints.
3. **Regulatorische Schranken** (`regulatory_risk`): Dependency on tariffs, export bans, environmental restrictions, or resource nationalism.
4. **ESG-Faktor** (`esg_risk`): Ecological degradation, carbon intensity of extraction, social criteria compliance, and human rights aspects.
5. **Herstellerkonzentration** (`producer_concentration`): Aggregated index mapping global production monopolies (e.g., Rare Earth processing hubs).
6. **Preisvolatilität** (`volatility`): History of sharp and speculative price fluctuations in major mercantile exchanges.

Model Integration & System Prompts

The agent uses the `gemini-3.1-pro-preview` model to assess global risk vectors.

System Instruction

Du bist der "Risk Agent" der CAPITAL-AI Plattform.
Deine Aufgabe ist es, Versorgungsrisiken, Länderrisiken und ESG-bezogene Schwachstellen entlang der globalen Rohstofflieferkette zu analysieren.
Gib ausschließlich ein valides JSON-Objekt zurück, das dem verlangten Schema entspricht.

JSON Schema Specification

```
{
  "type": "OBJECT",
  "properties": {
    "geopolitical_risk": { "type": "INTEGER", "description": "Geopolitisches Risiko von 0 bis 100" },
    "supply_chain_risk": { "type": "INTEGER", "description": "Lieferkettenrisiko von 0 bis 100" },
```

```
"regulatory_risk": { "type": "INTEGER", "description": "Regulatorisches Risiko von 0 bis 100" },
"esg_risk": { "type": "INTEGER", "description": "ESG/Umweltrisiko von 0 bis 100" },
"producer_concentration": { "type": "INTEGER", "description": "Herstellerkonzentration von 0 bis 100" },
"volatility": { "type": "INTEGER", "description": "Preisvolatilität von 0 bis 100" },
"explanation": { "type": "STRING", "description": "Kurze qualitative Zusammenfassung der Risikotreiber" }
},
"required": ["geopolitical_risk", "supply_chain_risk", "regulatory_risk", "esg_risk", "producer_concentration", "volatility", "explanation"]
}
```

Risk Mitigation Strategy

Risk scores are heavily weighted inside critical commodity configurations to calculate high-security cushions, warning industrial decision-makers of high country-of-origin concentrations.

Verified under CAPITAL-AI Platform Specification Version 0.5.4.

Scoring Model - CAPITAL-AI Raw Materials Engine

Quelle: docs/archive/raw-materials/Scoring-Model.md

Scoring Model - CAPITAL-AI Raw Materials Engine

This document outlines the mathematical models, dynamic weights, and formula implementations used inside the centralized CAPITAL-AI **Scoring Engine** (`RawMaterialsScoringService`).

Scoring Formula & Dimensions

The global score is computed on a scale of **0 to 100**, aggregated across **5 core dimensions**:

$$Score_{Final} = \sum_{d \in Dimensions} Score_d \times Weight_d$$

Where the active scoring configurations map as follows:

Dimension	Key Identifier	Description	Standard Weight (v1.0)	Critical Weight (v1.2)
Markt & Liquidität	liquidity	Volatilität, physisches Handelsvolumen, Börsenaktivität	15%	10%
Fundamentaldaten	fundamentals	Vorkommen, Reichweite, Erzgehalt, Substituierbarkeit, Recycling	25%	20%
Gewinnung & Komplexität	processing	Raffinationsaufwand, Infrastruktur, Förderkosten	15%	15%
Risiko & Resilienz	risk	Geopolitik, Lieferkette, Zölle, ESG-Konformität, Konzentration	25%	35%
Strategische Relevanz	strategicValue	Wehrtechnische und zukunftsindustrielle Bedeutung	20%	20%

Sub-Score Calculations

1. Markt & Liquidität

Calculates standard trading liquidity based on exchange listings and trading volumes: \$

$$\text{Score}_{\text{Liquidity}} = \frac{\text{Liquidity}_{\text{Market}} + \text{Volume}_{\text{Trading}}}{2}$$

2. Fundamentaldaten

Aggregates geological abundance, depletion rates, recyclability, and substitution difficulties:
$$\text{Score}_{\text{Fundamentals}} = \frac{\text{Grade}_{\text{Ore}}}{2} + \frac{\text{Tonnage} + \text{Tonnage}_{\text{Reserve}}}{2} + \text{Substitution}_{\text{Potential}} + \text{Recyclability}_{4}$$

3. Gewinnung & Komplexität

Assesses mechanical extraction friction. Lower complexity and costs yield higher positive scores:
$$\text{Score}_{\text{Processing}} = \frac{(100 - \text{Complexity}_{\text{Processing}}) + \text{Availability}_{\text{Infra}} + (100 - \text{Costs}_{\text{Extraction}})}{3}$$

4. Risiko & Resilienz

Aggregates supply chain bottlenecks. Low risks result in higher score contributions:
$$\text{Score}_{\text{Risk}} = \frac{\text{Risk}_{\text{Geopolitical}} + \text{Risk}_{\text{SupplyChain}} + \text{Risk}_{\text{Regulatory}} + \text{ESG}_{\text{Friction}} + \text{Concentration}_{\text{Producer}} + \text{Volatility}}{6}$$

$$\text{Contribution}_{\text{Risk}} = 100 - \text{Score}_{\text{Risk}}$$

5. Strategische Relevanz

Measures critical applications:
$$\text{Score}_{\text{Strategic}} = \frac{\text{Importance}_{\text{Military}} + \text{Importance}_{\text{Industrial}}}{2}$$

Missing Data & Confidence Penalty

When data points are missing, the engine gracefully handles them to prevent crashing, applying a penalty: - **Data Quality Rating:**
$$\text{Ratio}_{\text{Missing}} = \frac{\text{Missing Fields}}{\text{Total Fields (18)}}$$
 -
$$\text{Ratio} \leq 15\% \rightarrow \text{High quality}$$
 -
$$\text{Ratio} \leq 35\% \rightarrow \text{Medium quality}$$
 -
$$\text{Ratio} \leq 60\% \rightarrow \text{Low quality}$$
 -
$$\text{Ratio}_{\text{Missing}} > 60\% \rightarrow \text{Unknown quality}$$

- **Confidence Score:**
$$\text{Confidence} = \max(0.15, \text{BaseConfidence} - (\text{MissingCount} \times 0.04))$$

Verified under CAPITAL-AI Platform Specification Version 0.5.4.

Valuation Agent Documentation - CAPITAL-AI

Quelle: docs/archive/raw-materials/Valuation-Agent.md

Valuation Agent Documentation - CAPITAL-AI

*The **Valuation Agent** (`ValuationAgent`) focuses on the strategic importance of a raw material within international security architectures and high-tech technological roadmaps.

Functional Responsibilities

1. **Wehrtechnische Relevanz** (`military_importance`): Assess the indispensability of the material for defense equipment, aerospace engineering, hypersonic structures, and heavy armor steel.
2. **Industrielle Relevanz** (`industrial_importance`): Appraise the strategic value of the material within critical innovations, semiconductor manufacturing, grid infrastructure, and green technologies (e.g. electric mobility, solar panels).

Model Integration & System Prompts

The agent uses the `gemini-3.1-pro-preview` model to fetch and consolidate defense and industrial strategic dossiers.

System Instruction

Du bist der "Valuation Agent" der CAPITAL-AI Plattform.
Deine Aufgabe ist es, die wehrtechnische und gesamtindustrielle Relevanz von Rohstoffen entlang internationaler Sicherheits- und Innovationsstrategien einzustufen.
Gib ausschließlich ein valides JSON-Objekt zurück, das dem verlangten Schema entspricht.

JSON Schema Specification

```
{
  "type": "OBJECT",
  "properties": {
    "military_importance": { "type": "INTEGER", "description": "Militärische Bedeutung von 0 bis 100" },
    "industrial_importance": { "type": "INTEGER", "description": "Industrielle Bedeutung von 0 bis 100" },
    "explanation": { "type": "STRING", "description": "Qualitative Zusammenfassung der strategischen Relevanz" }
  },
  "required": ["military_importance", "industrial_importance", "explanation"]
}
```

Verified under CAPITAL-AI Platform Specification Version 0.5.4.

Archived Documentation

Quelle: docs/archive/README.md

Archived Documentation

Deutsch

Dieser Bereich enthält historische CAPITAL-AI-Dokumente, deren Evidence-Wert erhalten bleibt, die jedoch keine aktuelle Architektur-, Governance-, Security-, Compliance- oder API-Authority darstellen.

Kategorien: - `raw-materials/` : ältere Raw-Materials-Agent-, Scoring-, Orchestrator- und API-Snapshots; - `legacy/` : überholte Blueprint-, Architektur-, Deployment- und Integrationsdokumente; - `evidence/` : historische Security-/Compliance-Audit-Snapshots.

Aktuelle technische Entscheidungen müssen gegen ESS, ADRs, aktuelle `docs/architecture/` -Dokumente, Registries, Tests und Production Evidence geprüft werden.

English

This directory preserves historical CAPITAL-AI documentation for evidence and traceability only. Files under `docs/archive/` are non-canonical and must not override current ESS contracts, ADRs, architecture specifications, registries, tests or production evidence.